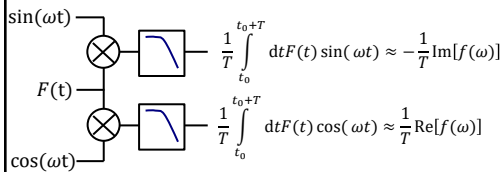


Swept-tuned spectrum analyzer in more detail



The input signal of the spectrum analyzer ($F(t)$) is multiplied by a signal of circular frequency ω ($\cos(\omega t)$) and its 90 degree shift ($\sin(\omega t)$). The multiplied signals are passed through low-pass filters. A low-pass filter is equivalent to integrating the (multiplied) signal over a certain time T . It can be seen that the two outputs are proportional to the real and the imaginary parts of the Fourier transform of the signal $F(t)$ at frequency ω . By varying the circular frequency ω , we can map the full Fourier spectrum.

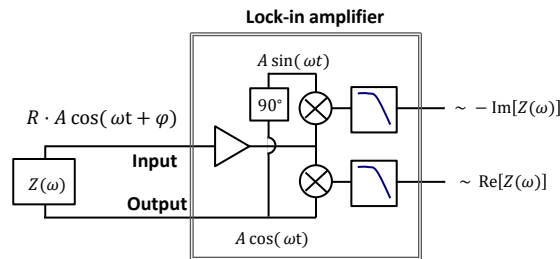
Lock-in amplifier

Practically the same thing is done by the so-called lock-in amplifier. The only difference is that $F(t)$ is not an arbitrary external signal, but a physical system is driven by the $\cos(\omega t)$ signal output by the lock-in and the response is measured at the input of the lock-in. For a linear system, the response function is a multiplication by a complex number ($Z(\omega)$), which can be described by an amplitude (R) and a phase shift (ϕ).

The lock in, after amplification, multiplies the signal at its input by the drive signal ($\sim \cos(\omega t)$) and its 90 degree shift ($\sim \sin(\omega t)$), passing the multiplied signals through a low-pass filter (for a digital lock in amplifier, this is effectively integration). The resulting signals are the real and the imaginary parts of the complex response function $Z(\omega)$. This method effectively filters out any interfering signal at a frequency other than that of the drive. This is called **phase-sensitive detection**.

$$\text{Drive: } \text{Re}[A \cdot e^{i\omega t}] = A \cdot \cos(\omega t)$$

$$\text{Response: } \text{Re}[Z(\omega) \cdot A \cdot e^{i\omega t}] = \text{Re}[Z(\omega)] \cdot A \cdot \cos(\omega t) - \text{Im}[Z(\omega)] \cdot A \cdot \sin(\omega t)$$



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Phase locked loop, PLL

Goal: to synchronize the output of the PLL to the input signal (this is also used in a lock-in, see next slide)

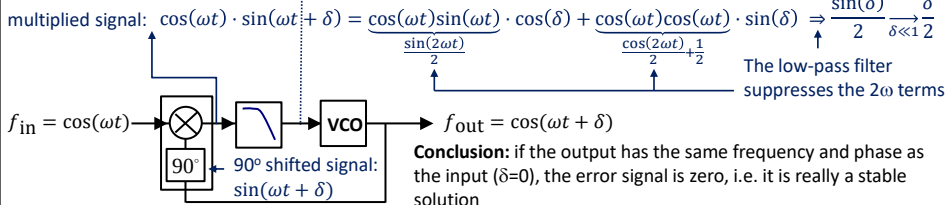
Phase detector: generates an error signal proportional to the phase difference between the output and input signal

Low-pass filter: filters the sum frequency from the multiplied signal

Voltage controlled oscillator: changes the output frequency with a speed proportional to the input voltage (i.e. the error signal), $df/dt \sim V_{in}$. This is done with a PI regulator (see temperature control in one of the first lectures) using a setpoint, where the error signal is zero.

Fixpoint: if the output and input signals are of the same frequency and phase, the phase detector gives a zero error signal, so the VCO does not change frequency. The output signal is then synchronized to the input signal. If the input signal is "noisy" around a given frequency, it will be "cleaned up" at the output.

Example:

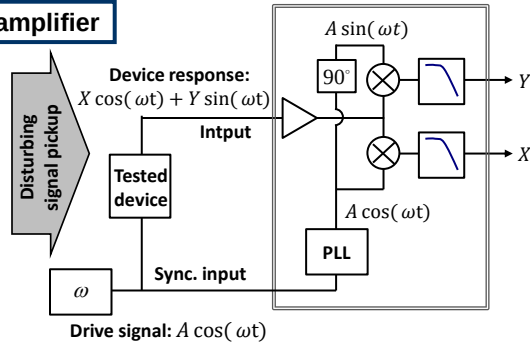


If, however, the output has the same frequency but different phase (nonzero constant δ), or moreover, the output is somewhat detuned in frequency (consider $\delta = \Delta\omega \cdot t + \delta_0$), the nonzero error signal is compensated by the PI loop of the PLL and the VCO will finally find the stable solution, where the output is "locked" to the input.

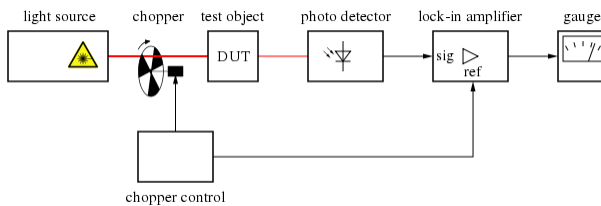
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PLL synchronization in a lock-in amplifier

The lock in amplifier can also be synchronized to an external signal. We drive the system with an ω circular frequency signal (not necessarily harmonic). This signal is also connected to the synchronizing input of the lock-in synchronizer. The lock-in uses a PLL to synchronize its internal oscillator to the drive signal and performs the multiplication by its own synchronized (harmonic) signal, or by shifting it by 90 degrees.



Measuring a small signal in a noisy environment using a modulation technique: we illuminate an object with a laser and want to measure the transmitted light with a photodetector. However, the detector can also "see" light coming from elsewhere. We modulate our signal with a frequency f (e.g. we chop it up), the lock-in amplifier is synchronized to the modulating signal, so that the lock-in input of the detector only sees the effect of the modulated laser light, everything else is "filtered out".



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Higher harmonic measurement with a lock-in

Higher harmonic generation in a nonlinear system:

The response of a nonlinear system to harmonic driving also includes higher harmonics. With a lock-in amplifier and a $\cos(\omega t)$ drive, we can measure the $\cos(n\omega t)$ and $\sin(n\omega t)$ response, i.e. the amplitude and phase of the n^{th} harmonic.

Example: consider a circuit element with nonlinear current-voltage ($I(V)$) characteristics. To a DC voltage V_0 , we add a low amplitude AC signal:

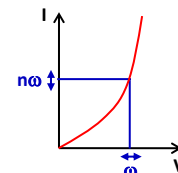
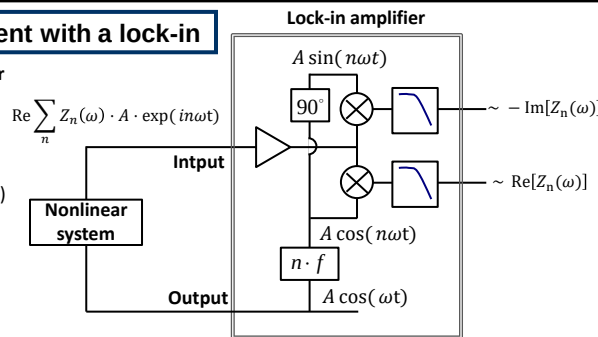
$$V(t) = V_0 + V_1 \cos(\omega t)$$

Measure the AC current with $n\omega$ circular frequency at the lock in input. What will be the amplitudes of the different harmonics?

$$I(V(t)) = I(V_0 + V_1 \cos(\omega t)) \approx I(V_0) + \left. \frac{dI}{dV} \right|_{V_0} \cdot V_1 \cos(\omega t) + \frac{1}{2} \left. \frac{d^2 I}{dV^2} \right|_{V_0} \cdot V_1^2 \cos^2(\omega t)$$

That is, the response at the fundamental frequency ($n=1$) will be proportional to the derivative of the $I(V)$ curve, and the second harmonic response ($n=2$) will be proportional to the second derivative of the $I(V)$ curve, the latter measuring the degree of nonlinearity.

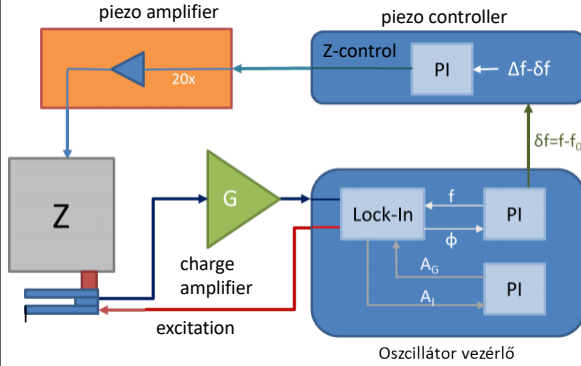
With this method, the first and second derivatives of the $I(V)$ curve can be measured directly while slowly varying V_0 , without numerically differentiating the $I(V)$ curve measured with a DC instrument.



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Atomic force microscope with a quartz tuning fork

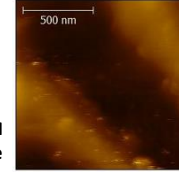
The "clock signal" in the watches is provided by a tuning-fork-shaped quartz oscillator, which can be excited electrically due to its piezoelectric behaviour. By sticking an STM needle on one of the branches of the tuning fork, not only the tunnel current but also the force between the sample and the needle can be studied by the resonant frequency of the quartz oscillator. Based on the force measurement, we can even take images of insulating surfaces. To do this, a complex control scheme has to be implemented:



During scanning, the resonance frequency is varied due to the force between the needle and the sample. The PLL tells the actual resonance frequency (δf) relative to the free resonance frequency (f_0). The z-direction piezo is moved by means of a third PI controller so that the frequency detuning, i.e. the force between the needle and the sample, is constant. This corresponds to a Δf setpoint detuning. (Precisely: the frequency detuning is not sensitive to the force but to the spring constant, i.e. the force gradient)



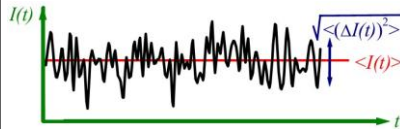
A PLL is used to excite the quartz oscillator continuously at its resonance frequency: i.e. a voltage with ω circular frequency and the current of the quartz oscillator is measured. A PI controller is used to control the frequency so that the phase shift is zero (i.e. it is excited at exactly the resonance frequency). Another PI controller is used to keep the amplitude of the oscillation constant.



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Spectral density of noise

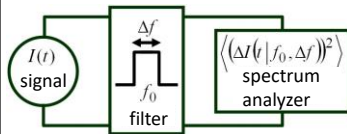
Usually, we are interested in the average value of a physical quantity (e.g. current), but around this average value we may experience fluctuations for various reasons:



The deviation of the current from the mean: $\Delta I = I(t) - \langle I(t) \rangle$

By definition: $\langle \Delta I(t) \rangle = 0$

Experimental definition of the spectral density of fluctuations: The signal $I(t)$ is passed through a band-pass filter with a narrow bandwidth Δf around a center frequency f_0 and the $\langle (\Delta I(t|f_0, \Delta f))^2 \rangle$ mean squared deviation of the such filtered signal is measured. The latter is proportional to the width of the frequency window Δf , and the proportionality factor is the $S_I(f_0)$ spectral density of the current noise at f_0 :



Spectral density of noise, or power spectrum density, PSD

Unit:

$$[S_I] = \frac{A^2}{Hz}$$

The mean squared deviation of the current over the full frequency range:

$$\langle (\Delta I(t))^2 \rangle = \int_0^\infty df S(f)$$

Similarly, for voltage noise:

$$\langle (\Delta V(t|f_0, \Delta f))^2 \rangle = S_V(f_0) \Delta f$$

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